

Solar Absorption Fridge

A solar-powered NH₃-H₂O absorption refrigerator for off-grid household cooling in Sudan. Thermodynamic model, CAD, P&ID, and Bill of Materials.

A solar fridge without a compressor

Absorption refrigeration replaces mechanical work with heat. The NH₃-H₂O pair, first used in the 1850s, is proven — but difficult to model with modern libraries.

200 _W COOLING	150 _L COLD BOX	0.424 COP	1.59 _{m2} COLLECTOR
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Purpose

Build a household-scale refrigerator that runs entirely on solar heat, with no mechanical compressor and no moving parts. Target: off-grid communities in Sudan where grid electricity is unreliable.

The four-component cycle

Solar heat drives ammonia out of a rich NH₃-H₂O solution in the **generator**. The vapor condenses in the **condenser** and flashes into the **evaporator**, where it absorbs heat from the cold box. It then returns to the **absorber** where poor solution from the generator reabsorbs it. No compressor — pressure difference is maintained by heat and gravity alone.

KEY DESIGN DECISION

NH₃-H₂O was chosen over LiBr-H₂O because it can reach -15 C. LiBr-H₂O cannot go below 0 C and requires vacuum seals.

Working around CoolProp's binary pair gap

CoolProp does not include a binary interaction pair for NH3-H2O. This project documents the gap and provides a validated alternative.

The gap

CoolProp's HEOS backend requires departure-function parameters for each binary pair. For NH3-H2O, those parameters have never been fitted. A direct call fails with:

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ValueError: Could not match the binary pair [7664-41-7, 7732-18-5]
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This is a data gap, not an implementation bug. The infrastructure exists but the fit has never been supplied. See CoolProp Issue #341, open since 2014.

Three working paths

PATH	ACCURACY	LICENSE
teqp (Tillner-Roth 1998)	Highest (IAPWS-2001)	Open source
Ziegler-Trepp 1984 (used here)	+/- 0.3-0.6 C	Open, published
REFPROP	Reference	Commercial

Validation

TEST	COMPUTED	REFERENCE	DELTA
Pure NH3 at 15.5 bar	39.70 C	40.0 C (CoolProp)	0.30 C
Pure NH3 at 10 bar	24.65 C	24.9 C (CoolProp)	0.25 C
Pure H2O at 1 bar	98.98 C	99.6 C (NIST)	0.62 C
y_NH3 at 15.5 bar, x=0.25	0.9059	0.90-0.92 (Herold)	in band

COEFFICIENT TRACEABILITY

All 16 excess-Gibbs coefficients (E1-E16) were cross-validated between Kherris et al. (2013) and Sadhukhan et al. with 100% agreement.

Mechanical specifications

All components sized for 200 W cooling, 150 L cold box, $T_{\text{evap}} = -15$ C, $T_{\text{cond}} = 40$ C, $T_{\text{gen}} = 135$ C.

COMPONENT	SPECIFICATION
Generator / receiver (V-101)	200 mm OD x 4 m, carbon steel, 2.5 mm wall, 48.5 kg
Solar collector (E-101)	CPC, 400 mm x 3980 mm, aperture 1.59 m2, acceptance 30 deg
Condenser (E-102)	7.5 m of 15 mm finned steel, air-cooled
Evaporator (E-104)	6.37 m of 10 mm finned steel, in cold box
Solution heat exchanger	70% effectiveness, 277 W duty
Absorber (V-102)	Integrated with generator; 3.97 m2 available
Rich tank (V-103)	3.0 L, 40% NH3
Poor tank (V-104)	2.3 L, 25% NH3

Operating conditions

PARAMETER	VALUE
Low-side pressure	2.36 bar (evap + absorber)
High-side pressure	15.55 bar (gen + condenser)
Refrigerant flow	0.68 kg/h
Rich solution flow	2.99 kg/h
Poor solution flow	2.30 kg/h
Circulation ratio	4.37

Procurement summary

35 items across 8 categories. Total ~2,184 USD (1.31M SDG at Sep 2026 rate).

CATEGORY	USD	SHARE
Steel pipes and vessels	525	24.0%
Fabrication services	390	17.9%
Solar collector	304	13.9%
Valves	283	13.0%
Insulation and cold box	280	12.8%
Instruments	152	7.0%
Consumables	145	6.6%
Working fluids	105	4.8%
Total	2,184	100%

SUDAN PROCUREMENT NOTE

Steel pipes: Khartoum Bahri Industrial Area or Port Sudan. TIG welding: workshops in Khartoum industrial area, ~20-30 USD/hr. Ammonia: agricultural fertilizer suppliers, request 99.5% purity. Safety valves: import genuine parts — do not compromise.

Design-complete, prototype pending

PHASE	STATUS
Thermodynamic model	Complete (validated +/- 0.3 C)
Cycle simulation	Complete (COP 0.424)
Mechanical design	Complete (all components sized)
CAD models	Complete (FreeCAD scripts + STL)
P&ID	Complete (matplotlib + draw.io)
Bill of Materials	Complete (35 items)
Prototype	Not yet built
Experimental validation	Pending prototype

References

Ziegler & Trepp (1984), Int. J. Refrigeration 7(2):101-106. Patek & Klomfar (1995), Int. J. Refrigeration 18(4):228-234. Tillner-Roth & Friend (1998), J. Phys. Chem. Ref. Data 27(1):63-96. Kherris et al. (2013), Thermal Science 17(3):891-902. Herold, Radermacher, Klein, Absorption Chillers and Heat Pumps, CRC Press.